Unit 3 - Session 8

SLO 1: GHT for Feature Collation & Advanced Topics-

Synthesize partial evidence using probabilistic voting

Duration: 10 Minutes Activity

Question:

Fill in the Blanks

Q1	In standard GHT, every boundary point casts a vote of exactly **1** into the **accumulator**, regardless of how strong or reliable the edge is.
Q2	In **Probabilistic GHT**, each boundary point is assigned a **confidence score** between 0.0 and 1.0, which determines how much it contributes to the vote.
Q3	A stronger edge with a higher **gradient magnitude** will receive a higher **weighted vote**, meaning it influences the accumulator peak more than a weak, noisy edge.
Q4	When part of an object is hidden from view, only some boundary points are visible. This situation is called **occlusion**, and the reduced set of votes is known as **partial evidence**.
Q5	A **false peak** is a spurious accumulator peak caused by unrelated edge votes. A detection **threshold** is used to decide the minimum score a peak must reach before it is accepted as a valid object detection.

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Match the Column [10 Marks | 1 mark each]

Match each term in Column A with its correct definition in Column B. Write the number of the matching definition in the Answer column.

	COLUMN A — Term	Answer (write number)		COLUMN B — Definition	No.
A	Weighted Vote	3	5	The location in the accumulator with the highest total weighted votes, indicating the most likely object pose.	5
B	Accumulator Peak	5	8	A measure of edge strength at a boundary point; higher values indicate cleaner, more reliable edges.	8
C	Confidence Score	2	1	The multi-dimensional voting space where each bin stores the total vote score for a parameter combination.	1
D	Detection Threshold	7	10	The physical blocking of part of an object by another object, preventing some boundary points from being detected.	10
E	False Peak	4	3	The vote contribution of a single boundary point, equal to its confidence score rather than a fixed value of 1.	3
F	Partial Evidence	6	6	The condition where only a subset of expected boundary votes is cast, typically due to occlusion or noise.	6
G	Gradient Magnitude	8	9	A variant of the Generalized Hough Transform where votes are weighted by confidence rather than being uniform.	9
H	Probabilistic GHT	9	2	A value between 0.0 and 1.0 assigned to a boundary point based on the reliability of its edge gradient.	2
I	Occlusion	10	7	The minimum accumulated vote score that a peak must exceed before it is accepted as a valid object detection.	7
J	Accumulator	1	4	A spurious high-vote region in the accumulator caused by unrelated or coincidental edge alignments.	4